EC442 Macroeconomics for MRes students | Instructor: Matthias Doepke
London School of Economics | Autumn 2026

Problem Set 2

Dynamic Programming: Computing Numerical Solutions

Question 1: Consider the Bellman equation for the social planning problem for the "standard model:"

$$v(k) = \max_{k'} \left\{ \log\left((1-\delta)k + Ak^{\alpha} - k'\right) + \beta v(k') \right\}.$$

Compute the value function and the policy function (the optimal choice of k' given k) using value function iteration on a discrete grid. Use the parameter values $\beta = 0.9$, $A = 3$, $\alpha = 0.3$, and $\delta = 0.2$. You can use $\bar{k} = 10$ as the upper bound for capital. Use $v_0(k) = 0$ for all k as your initial guess. Your capital grid should have at least 10,000 grid points.

1. Provide plots of the value and policy functions, and determine the steady-state level of capital.
2. Compute and plot time series for capital over 20 periods with starting values of $k_0 = 0.1$, $k_0 = 1$, and $k_0 = 10$.
3. Comment on the implications of your results for the convergence of income levels across countries.

Question 2: Redo your computation of the value function using the Howard policy improvement algorithm described in Ljungqvist and Sargent. Briefly, this algorithm also iterates on the value function, but the policy function is only optimized infrequently (after the value function for a given policy has converged) rather than optimizing at every iteration. Use the same grids as above, use the initial guess $v_0(k) = 0$ for all k, and construct an optimal policy $k' = g_0(k)$ for all k corresponding to this value function (this is what you did in the first iteration above). Now, before optimizing the policy again, iterate on the value function with the *fixed* policy function $g_0(k)$ until convergence, using the criterion 10^{-5}. Call the resulting value function associated with this policy $v_1(k)$. Then, compute a new policy using $v_1(k)$ as the value function from tomorrow on. Compute the value $v_2(k)$ associated with this fixed policy function by iteration, taking $v_1(k)$ as the initial guess. Continue in this fashion until convergence of the value function. Verify that this algorithm gives the same outcome as standard value function iteration, and compare the required computation time for each algorithm. What explains the difference in computation time?

Dynamic Programming: Time to Build

Question 3: Consider the following economy. The representative consumer has preferences:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t).$$

Output is produced using physical capital k_t. Current output can be transformed into consumption c_t or into investment i_t. It takes two periods for current investment to be transformed into productive capital. The resource constraints for the economy are therefore given by:

$$\begin{aligned} c_t + i_t &= Ak_t, \\ k_{t+2} &= i_t. \end{aligned}$$

The initial capital stocks for the first two periods k_0 and k_1 are given.

1. Formulate the social planning problem for this economy in sequence form.
2. Provide a Bellman equation that corresponds to the planning problem.
3. Conjecture that the value function is log-linear and that the policy function is linear, and solve for the value and policy functions.
4. Compute the balanced growth rate of the economy as a function of parameters. Does the one-period consumption growth rate c_{t+1}/c_t converge to a constant from any initial condition?

A Two-Sector Economy

Question 4: Consider an economy with a representative consumer whose preferences are described by:

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}.$$

Assume that there are separate technologies for producing the consumption good and the investment good. Capital and labor can be moved freely across the two sectors. The resource constraints are given by:

$$\begin{aligned} c_t &\leq l_{ct}^{\alpha} k_{ct}^{1-\alpha}, \\ i_t &\leq l_{it}^{\gamma} k_{it}^{1-\gamma}, \\ l_{it} + l_{ct} &\leq 1, \\ k_{it} + k_{ct} &\leq k_t, \\ k_{t+1} &= (1-\delta)k_t + i_t, \end{aligned}$$

with k_0 given.

1. Write down the sequential planning problem for this economy. What is the state variable?

2. Write down the Bellman equation for the recursive formulation of the planning problem.

3. Define a state space X that ensures that the utility function is bounded on the state space, and show that the operator defined by the Bellman equation satisfies Blackwell's sufficient conditions for a contraction.

4. Assume that the value function is differentiable, and derive and interpret the first-order and envelope conditions that characterize the optimal solution.

5. Show how you can use the conditions to find the steady state of the economy.

Sequential-Markets Equilibria and Arrow-Debreu Equilibria

Question 5: Consider the standard optimal growth model represented by the planning problem:

$$\max_{\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \right\}.$$

subject to:

$$\begin{aligned} c_t + k_{t+1} &\leq (1-\delta)k_t + A(k_t)^{\alpha}(n_t)^{1-\alpha}, \\ 0 \leq n_t &\leq 1, \end{aligned}$$

with k_0 given.

1. Define a sequential-markets equilibrium.

2. Define an Arrow-Debreu equilibrium.

3. Show that the two types of equilibria are equivalent. That is, for a given sequential-markets equilibrium, find the prices that implement the equilibrium allocation as an Arrow-Debreu equilibrium, and show that all equilibrium conditions are satisfied. Follow the same approach to show that any Arrow-Debreu equilibrium can be transformed into a sequential-markets equilibrium.